



# SIMATS Online Programming Lab

## DS PROGRAMMING



YUGITHA B  
192424102RD



### QUESTIONS

2 / 10 completed

Queue Using Arrays

### QUEUE USING ARRAYS

Write a C program to implement a Queue using Arrays with operations such as insertion (enqueue), deletion (dequeue), display, and peek. The program should simulate a ticket booking counter where customers arrive in sequence and are served on a first-come-first-served basis. The queue should maintain the order of customer requests and display the current status after each operation. Sample Input: Enqueue 101, Enqueue 102, Enqueue 103, Display, Dequeue, Display. Sample Output: Queue Elements: 101 102 103. Deleted Element: 101. Queue Elements: 102 103.

**PROGRAM EDITOR**

C

Run

Save

```
1 #include <stdio.h>
2 #define MAX 5
3
4 int queue[MAX];
5 int front = -1;
6 int rear = -1;
7
8 void enqueue(int value) {
9     if (rear == MAX - 1) {
10         printf("Queue Overflow!\n");
11     } else {
12         if (front == -1) front = 0;
13         rear++;
14         queue[rear] = value;
15     }
16 }
17
18 void dequeue() {
19     if (front == -1 || front > rear) {
20         printf("Queue Underflow!\n");
21     } else {
```

**OUTPUT**

```
Queue Elements: 101 102 103
Deleted Element: 101
Queue Elements: 102 103
```

**INPUT**

Enqueue 101, Enqueue 102, Enqueue 103, Display, Dequeue, Display, Dequeue, Display

© 2026 **SIMATS Online Lab**. All rights reserved.